

AUCTION INSIGHT AI

Software Requirements Specification

AI-Powered Japanese Vehicle Auction Sheet Analysis Platform

Field	Value
Version	1.0.0
Status	Draft for Review
Primary Market	Sri Lanka (LKR) — Extensible Globally
Date	May 2026
Classification	Confidential — Investor & Engineering Use

1. Executive Summary

Auction Insight AI is a cloud-native, AI-powered Software-as-a-Service (SaaS) platform that transforms raw Japanese vehicle auction sheets into structured, actionable vehicle intelligence reports. The platform targets vehicle importers, car dealers, auction agents, and individual buyers operating in right-hand-drive import markets, with Sri Lanka as the primary launch market.

Japanese auction houses — including USS, JAA, JU, TAA, and regional networks — issue standardised condition sheets written almost entirely in Japanese. These sheets encode vehicle grade, mileage, chassis number, equipment, damage notation, inspection comments, and repair history using a proprietary grading vocabulary and pictographic damage diagrams. Currently, importers rely on human translators, bilingual agents, or partial machine-translation services that miss domain-specific damage codes, producing unreliable assessments and significant financial risk.

Auction Insight AI solves this problem end-to-end: a user uploads a PDF or image of an auction sheet; the platform runs a multi-stage AI pipeline (OCR → translation → structured extraction → damage interpretation → report generation) and delivers a human-readable, professionally formatted vehicle intelligence report within seconds.

Platform Highlights

- Sub-30-second end-to-end analysis latency (Phase 1 target)
- Support for all major Japanese auction sheet formats: USS, JAA, JU, TAA
- Pay-per-analysis and subscription billing models in LKR and USD
- Multi-tenant architecture enabling white-label dealer portals
- Phased delivery over five release increments spanning 18 months
- Built on AWS, Java Spring Boot, Python FastAPI, and React / Three.js

Business Model

Revenue Stream	Unit Price	Target Volume
Pay-per-analysis (Standard)	LKR 250	1,500/month
Pay-per-analysis (Priority)	LKR 500	300/month
Dealer Starter Pack (10 credits)	LKR 2,000	50 packs/month
Dealer Basic Subscription	LKR 12,000/month	10 tenants
Dealer Pro Subscription	LKR 25,000/month	5 tenants

2. Business Problem

2.1 Market Context

Sri Lanka imports approximately 40,000–60,000 used vehicles annually, the majority sourced from Japanese auction houses. The import chain involves auction agents in Japan who bid on behalf of Sri Lankan buyers, shipping agents, customs brokers, and domestic dealers. The condition and price of a vehicle are directly correlated to the information encoded on its auction sheet.

2.2 Core Pain Points

Language Barrier

Auction sheets are printed almost entirely in Japanese — grading comments, damage descriptions, inspection remarks, and special notes are inaccessible to the majority of Sri Lankan importers without a Japanese-speaking intermediary.

Grading Misinterpretation

Japanese auction houses use proprietary grading systems (Grade 6, 5, 4.5, 4, 3.5, 3, 2, 1, RA, R) and damage codes (E1, E2, W1, W2, A1, U) that are not intuitive. A grade 3.5 vehicle with W2 (windscreen replacement) and U1 (undercarriage rust) represents a materially different risk profile from a 3.5 with only cosmetic scratches.

Damage Visualisation Gap

Auction sheets contain 2D top-down/side-view vehicle diagrams with handwritten damage notation markers. Currently, these are assessed by eye with no digital overlay, quantification, or comparative history.

Slow Turnaround

Manual translation and assessment takes 2–24 hours depending on agent availability. In live auction environments (USS holds 3–4 auctions per week), buyers must decide within minutes, making slow turnaround commercially unacceptable.

No Historical Intelligence

Importers have no structured database of past analyses, making it impossible to track patterns, benchmark auction grades against real condition, or assess agents' accuracy over time.

Fraud Risk

Odometer tampering, chassis number misrepresentation, and accident concealment are documented risks. A structured AI analysis creates a timestamped, auditable record that can flag inconsistencies.

2.3 Competitive Gap

Existing Solution	Limitation
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Generic OCR tools	No domain understanding of auction vocabulary
Google Translate	Misses grading codes, damage notation, auction-specific terms
Manual bilingual agents	Slow, expensive, inconsistent, no audit trail
Japanese auction portals	Requires Japanese literacy; access restricted to registered agents

3. Product Vision

3.1 Vision Statement

"To be the intelligence layer between every Japanese auction sheet and every import decision — globally."

3.2 Product Principles

- Speed over perfection in Phase 1 — a fast, good report beats a slow, perfect one in an auction environment
- Transparency of AI confidence — every extracted field carries a confidence score; ambiguity is surfaced, not hidden
- Domain depth over general AI — platform trains on auction-specific vocabulary, not generic NLP models
- Dealer-first distribution — white-label dealer portals are the primary growth vector
- Data as a moat — every analysis enriches the knowledge base, compounding accuracy and defensibility

3.3 Success Metrics (12-Month Targets)

Metric	Target
Analyses processed per month	3,000+
Report generation latency (p95)	< 35 seconds
Field extraction accuracy	> 92%
Paying dealer tenants	20+
Monthly Recurring Revenue	LKR 500,000+
User satisfaction (CSAT)	> 4.2 / 5.0
Platform uptime SLA	99.5%

4. User Personas

4.1 Persona A — The Independent Importer (Primary)

Attribute	Detail
Name / Profile	Pradeep, 38, Colombo
Role	Independent vehicle importer; 5–15 vehicles/month via Japan agent
Pain	Relies on agent WhatsApp summaries; deceived twice on condition — LKR 400,000+ in repairs
Needs	Self-serve English analysis, fast enough for bid decisions; damage visualisation
Tech literacy	Moderate — comfortable with web apps
Willingness to pay	LKR 200–300 per analysis; pay-as-you-go preferred

4.2 Persona B — The Car Dealer (High Value)

Attribute	Detail
Name / Profile	Sanjeewa, 45, Negombo — 3-franchise dealership
Role	30–50 vehicles/month; 2 staff assessing auction sheets
Pain	Inconsistent staff assessment quality; no audit trail; slow analysis loses bids
Needs	Team shared dashboard; bulk upload; branded reports for customers
Willingness to pay	LKR 15,000–20,000/month subscription

4.3 Persona C — The Auction Agent

Attribute	Detail
Name / Profile	Kenji, 34, Japan-based — bids for 20+ Sri Lankan clients
Pain	3–4 hours/day translating and summarising sheets via email/WhatsApp
Needs	API access for programmatic report generation; white-label branding
Willingness to pay	API subscription LKR 25,000+/month

5. Functional Requirements

5.1 Phase 1 — Core Analysis Engine

FR-001: Authentication

- Google OAuth 2.0 and email/password registration
- Email verification required before first analysis
- JWT RS256 tokens; 24-hour access token, 30-day refresh
- Rate-limited login: 5 attempts per 10 minutes per IP

FR-002: File Upload

- Accept PDF, JPEG, PNG, WEBP, HEIC up to 20 MB
- MIME type validation via magic byte sniffing
- AWS S3 storage with AES-256 server-side encryption
- Upload progress indicator and thumbnail preview

FR-003: OCR Extraction

- PaddleOCR Japanese model extracts all text with bounding boxes
- Auto-deskew images up to $\pm 15^\circ$ rotation
- Per-field confidence scores (0.0–1.0); flag < 0.75 for review
- Acceptance: CER $< 8\%$; extraction < 10 seconds

FR-004: Translation

- OpenAI GPT-4o with domain-specific system prompts
- Custom glossary of 500+ auction-domain Japanese terms
- Ambiguous translations flagged with alternative interpretations

FR-005: Structured Data Extraction

Canonical fields extracted for every sheet:

Field Group	Fields
Identity	Lot number, chassis/VIN, model code, model name
Dates	Registration date, manufacture date, inspection date
Specifications	Displacement (cc), transmission, drive type, doors, fuel type, seat capacity
Condition	Overall grade, interior grade, mileage
Appearance	Original colour code, changed colour
Dimensions	Length × Width × Height (cm)
Assessment	Good selling points, bad points, comments, inspector notes

FR-006: Damage Code Interpretation

Parses all standard USS/JAA damage codes:

Code Type	Values & Meanings
Location	A1=front bumper, A2=bonnet, A3=roof, A4=boot, A5=rear bumper, B1-B4=doors, C1-C4=panels
Damage Type	S=Scratch, D=Dent, W=Wave, C=Crack, X=Needs repair, XX=Severe, U=Rust, E=Replaced, P=Paint, B=Broken
Severity	1=Minor, 2=Moderate, 3=Severe

FR-007: Payment

- PayHere (LKR) primary; Stripe (international) fallback
- Pay-per-analysis and monthly subscription plans
- PDF receipt for every transaction; idempotency key prevents double-charging

FR-008: Dashboard

- Tabs: Overview, Condition, Damage, Comments, Raw Text
- Status filters; date range / model / grade filters

FR-009: Report Generation

- Branded PDF generated in < 5 seconds; stored 90 days
- Includes: vehicle overview, spec table, grading explanation, damage summary, AI confidence indicators, disclaimer

5.2–5.5 Phases 2–5 Feature Summary

Phase	Key Features
Phase 2	2D damage diagram overlay; Three.js 3D vehicle model with damage zones; vehicle type detection
Phase 3	Multi-sheet comparison (up to 4); AI risk summary (Low/Moderate/High/Do Not Buy); PDF/Excel/CSV export; history tracking
Phase 4	Multi-tenant architecture; white-label dealer portals; public REST API with API keys; subscription billing
Phase 5	RAG pipeline; AI chat assistant; quantitative risk scoring (0–100); predictive price analytics

6. Non-Functional Requirements

6.1 Performance Targets

Requirement	Target
End-to-end analysis (p50)	< 20 seconds
End-to-end analysis (p95)	< 35 seconds
API response time (status check)	< 300 ms
Dashboard page load	< 2.5 seconds
PDF report generation	< 5 seconds
Concurrent analyses (Phase 1)	50
Concurrent analyses (Phase 4)	500

6.2 Availability & Reliability

- Phase 1 SLA: 99.5% monthly uptime
- Phase 4 SLA: 99.9% monthly uptime
- RTO: 4 hours; RPO: 1 hour
- All AI pipeline steps idempotent; max 3 retries with exponential back-off
- Maintenance window: Sundays 00:00–02:00 LKT

6.3 Security

- AES-256 encryption at rest; TLS 1.3 in transit
- OWASP Top 10 compliance; annual penetration test
- Sri Lanka PDPA 2022 compliance
- Uploaded sheets scoped to uploading user/tenant — never shared cross-tenant

6.4 Usability

- WCAG 2.1 AA accessibility compliance
- Mobile-responsive down to 375px viewport
- First-time user completes analysis without documentation in < 5 minutes

6.5 Internationalisation

- Phase 1: English UI; Phase 2: Sinhala option
- Currency: LKR default; USD toggle for international

- Date format: DD/MM/YYYY (Sri Lanka standard)

7. System Architecture

7.1 Architecture Overview

The platform follows a microservices architecture deployed on AWS ECS Fargate, with a React SPA frontend served via CloudFront. The AI processing pipeline is decoupled from the Core API via AWS SQS, allowing burst absorption and independent scaling of compute-intensive AI workloads.

7.2 Microservice Inventory

Service	Technology	Primary Responsibilities
auth-service	Spring Boot	Registration, login, JWT issuance, Google OAuth
core-api-service	Spring Boot	Upload, user management, billing, reports, tenant
ai-orchestrator	Python FastAPI	Pipeline coordination, SQS consumer, status management
ocr-microservice	Python FastAPI	PaddleOCR, image pre-processing, bounding boxes
translation-service	Python FastAPI	GPT-4o translation, glossary lookup
extraction-service	Python FastAPI	LLM structured data extraction, JSON mapping
damage-service	Python FastAPI	Damage code parsing, manifest generation
report-service	Python FastAPI	WeasyPrint PDF generation, template rendering
admin-service	Spring Boot	Admin portal backend, audit logging
notification-service	Spring Boot	Email (SES), in-app, WebSocket notifications

7.3 Communication Patterns

Service Pair	Protocol	Pattern
Frontend to API Gateway	HTTPS REST	Request/Response
Core API to AI Orchestrator	AWS SQS	Async queue dispatch
AI Orchestrator to Client	WebSocket	Real-time status push
AI pipeline internal stages	HTTP REST	Sequential chained calls
Services to PostgreSQL	TCP/5432	HikariCP connection pool
Services to Redis	TCP/6379	Cache-aside pattern

8. AI Pipeline Architecture

8.1 Pipeline Stages

Stage	Service	Input	Output
1. Pre-processing	ocr-microservice	Raw image/PDF from S3	Deskewed, enhanced image
2. OCR Extraction	ocr-microservice / PaddleOCR	Pre-processed image	Text + bounding boxes + confidence
3. Region Classification	ocr-microservice	OCR results	Fields grouped by region
4. Glossary Lookup	translation-service	Classified fields	Deterministic translations
5. LLM Translation	translation-service / GPT-4o	Unmapped terms	English translations
6. Structured Extraction	extraction-service / GPT-4o	Translated fields	Canonical JSON schema
7. Damage Interpretation	damage-service / GPT-4o	Damage codes + context	Structured damage manifest
8. Confidence Scoring	ai-orchestrator	All field results	Per-field confidence scores
9. Report Assembly	report-service	Structured data + damage	HTML rendered from Jinja2
10. PDF Generation	report-service / WeasyPrint	HTML template	PDF bytes → S3

8.2 Error & Retry Strategy

Each pipeline stage is wrapped in a retry handler with exponential back-off. Maximum 3 retries per stage. After exhausting retries, the analysis is marked failed, the original file is preserved, and the user receives a notification with the option to re-upload a higher-quality image.

```
Retry delay: min(2^retry_count + random_jitter_seconds, max_delay_seconds)
Stage max delays: OCR=30s, Translation=60s, Extraction=60s, OpenAI=120s
```

9. OCR Pipeline Design

9.1 Pre-processing Algorithm

The pre-processing pipeline applies the following transformations in sequence before OCR:

- Greyscale conversion
- Skew detection and rotation correction (up to $\pm 15^\circ$) using minAreaRect
- Adaptive Gaussian thresholding to handle uneven lighting
- Fast non-local means denoising
- Unsharp masking for text edge sharpening

9.2 OCR Extraction Example

Sample structured output from PaddleOCR on a Nissan Skyline ECR32 auction sheet:

Raw OCR result (selected fields):

```
{ "text": "ECR32",      "confidence": 0.99, "region": "model_code" }
{ "text": "25GTS",     "confidence": 0.96, "region": "grade_trim" }
{ "text": "2500cc",    "confidence": 0.98, "region": "displacement" }
{ "text": "88278",     "confidence": 0.95, "region": "mileage_value" }
{ "text": "KH2",       "confidence": 0.89, "region": "colour_code" }
{ "text": "H4/2",      "confidence": 0.91, "region": "reg_date" }
{ "text": "ECR32-010999", "confidence": 0.99, "region": "chassis_number" }
```

9.3 Region-of-Interest Classification

Region	Y-range (% height)	X-range (% width)	Key Anchor Terms
Header / Model	0–15%	Full width	Vehicle name, model code
Specification block	10–35%	Full width	Mileage, cc, doors, transmission
Grade block	5–20%	Right 50%	Grade, interior
Damage diagram	50–95%	Right 50%	(Diagram region, no text anchor)
Comments / bad points	35–85%	Left 50%	Free-text Japanese comments

10. Translation Pipeline

10.1 Two-Stage Translation Logic

Stage 1 (Glossary Lookup): All OCR-extracted terms are checked against the curated domain glossary (500+ entries). Matches return confidence=0.99 without LLM call. This handles the majority of field labels, grades, and equipment codes.

Stage 2 (LLM Translation): Unmapped terms, free-text comments, and inspector notes are passed to GPT-4o with the translation system prompt. The LLM returns the translation plus an optional alternative interpretation for ambiguous terms.

10.2 Domain Glossary — Sample Entries

Japanese Term	English Translation	Category
Hyouka-ten (grade characters)	Overall Auction Grade	Grade
Naisou (interior characters)	Interior Grade	Grade
Soukou (mileage characters)	Mileage / Odometer	Specification
Shuufukureki (repair history chars)	Accident / Repair History (declared)	Condition
Wan-oonaa (one owner chars)	One Owner	Condition
Kinenshakan (non-smoking chars)	Non-smoking Vehicle	Condition
Kirokusho (record book chars)	Service Record Book Available	Documents
Sanruufu (sunroof chars)	Sunroof	Equipment
Kiireesu (keyless chars)	Keyless Entry	Equipment
AAC	Auto Air Conditioning	Equipment

11. Prompt Engineering Strategy

11.1 Translation Prompt Design Principles

- System prompt establishes domain expert persona (Japanese automotive auction specialist)
- Canonical term mappings embedded to override generic translations
- Japanese era date conversion rules (Heisei H → +1988, Reiwa R → +2018)
- Ambiguity handling: best translation followed by '(alt: alternative)' notation
- Strict output format: translation only, no commentary

11.2 Extraction Prompt Design Principles

- JSON Schema provided as strict output contract
- Null-over-guess rule: absent or illegible fields return null, not hallucinated values
- Mileage normalisation: miles-to-km conversion applied when unit detected
- Repair history: only set has_repair_history=true on explicit declaration
- Single JSON object only — no markdown fences, no explanation

11.3 Damage Interpretation Prompt Design Principles

- Full USS/JAA/JU damage code reference table embedded in prompt
- Multiple W codes on adjacent panels → flag possible prior accident in summary
- XX codes always trigger structural_concern=true
- E codes (replaced parts) noted as significant for valuation
- aggregate_severity_score: 0=pristine to 100=total loss — calibrated against USS norms

11.4 Prompt Version Management

All prompt versions are stored in the glossary_terms table with category='prompt_version'. Changes are reviewed by the Solution Architect, tested against the regression fixture set (minimum 95% accuracy maintained), and deployed via the admin Glossary Manager. Rollback is available to any previous version.

12. RAG Architecture (Phase 5)

12.1 Knowledge Base Structure

Category	Content	Chunk Strategy
auction_reports	Embedded historical analysis JSON results	Per-field chunks
damage_code_reference	Full USS/JAA/JU code vocabulary	Per-code entry
vehicle_model_data	JDM model specifications and known issues	Per-model entry
grading_standards	Auction house grading criteria and explanations	Per-grade entry
repair_cost_guides	Sri Lanka market repair benchmarks	Per-damage-type entry

12.2 Vector Database Schema

```
CREATE TABLE rag_embeddings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id UUID,
  source_type VARCHAR(50) NOT NULL,
  chunk_text TEXT NOT NULL,
  embedding vector(1536),          -- OpenAI text-embedding-3-small
  metadata JSONB,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX ON rag_embeddings USING ivfflat (embedding vector_cosine_ops) WITH
(lists = 100);
```

12.3 Query Flow

- User query embedded using text-embedding-3-small
- pgvector cosine similarity search returns top-8 relevant chunks
- Tenant-scoped search (own analyses + shared reference data)
- Context assembled and passed to GPT-4o with grounding prompt
- Response grounded in retrieved evidence; citations available on request

13. Database Design

13.1 Entity Relationship Summary

Table	Purpose	Key Relationships
tenants	SaaS tenant accounts	Parent of users, analyses, subscriptions
users	Platform users	Belongs to tenant; owns analyses and payments
analyses	Per-sheet analysis record	Belongs to user and tenant; stores JSONB results
payments	Transaction records	Belongs to user; linked to analysis or subscription
subscriptions	Monthly plan subscriptions	Belongs to user and tenant
glossary_terms	Domain translation vocabulary	Referenced by translation service
api_keys	Programmatic access keys (Phase 4)	Belongs to user and tenant
audit_log	Immutable event log	References user, tenant, entity
rag_embeddings	Vector chunks for RAG (Phase 5)	Scoped by tenant and source type

13.2 Analyses Table — Key Fields

Column	Type	Description
id	UUID	Primary key
tenant_id	UUID	Tenant scope FK
user_id	UUID	Owner FK
status	VARCHAR(50)	queued processing complete failed review_required
structured_data	JSONB	Full extracted vehicle data (canonical schema)
damage_manifest	JSONB	Parsed damage items and severity score
confidence_scores	JSONB	Per-field confidence values
overall_confidence	DECIMAL(4,3)	Weighted average confidence
report_pdf_s3_key	TEXT	S3 path to generated PDF report
retry_count	INTEGER	Number of pipeline retry attempts

13.3 Redis Cache Key Design

Key Pattern	TTL	Purpose
analysis:{id}:status	300s	Analysis status polling cache
user:{id}:credits	3600s	Credit balance cache
tenant:{id}:branding	86400s	White-label configuration
glossary:hash:v2	86400s	Full glossary lookup table
rate_limit:{ip}:login	600s	Brute-force login protection
session:{jti}	86400s	JWT session metadata

14. API Specifications

14.1 Base URL & Versioning

```

Production: https://api.auctioninsight.lk/v1
Staging:    https://api-staging.auctioninsight.lk/v1
White-label: https://api.{tenant-domain}/v1

```

14.2 Authentication Headers

```

Authorization: Bearer {jwt_access_token}
X-Tenant-ID:   {tenant_uuid}      (multi-tenant)
X-API-Key:     {api_key}          (public API, Phase 4)

```

14.3 Core Endpoint Reference

Method	Endpoint	Description	Auth
POST	/v1/analyses	Upload auction sheet; returns analysis_id + WebSocket URL	JWT
GET	/v1/analyses/{id}	Get analysis status and vehicle summary	JWT
GET	/v1/analyses/{id}/data	Get full structured vehicle + damage + confidence JSON	JWT
GET	/v1/analyses/{id}/report	Download PDF report (binary)	JWT
GET	/v1/analyses	List user's analyses with pagination and filters	JWT
POST	/v1/payments/orders	Create payment order (PayHere or Stripe)	JWT
POST	/v1/payments/payhere/notify	PayHere IPN webhook (server-to-server)	Hash
GET	/v1/users/me	Get current user profile and credit balance	JWT
DELETE	/v1/users/me/data	PDPA data erasure request	JWT
GET	/v1/admin/analyses	Admin: view all analyses across tenants	Admin JWT

14.4 Upload Response Example

```

POST /v1/analyses → 202 Accepted
{

```

```
"analysis_id": "550e8400-e29b-41d4-a716-446655440000",
"status": "queued",
"estimated_completion_seconds": 25,
"credits_deducted": 1,
"credits_remaining": 14,
"ws_channel": "wss://api.auctioninsight.lk/ws/analysis/550e8400"
}
```

14.5 Full Result — Structured Data Example

```
"vehicle": {
  "chassis_number": "ECR32-010999",
  "model_name": "Skyline",
  "model_code": "ECR32",
  "manufacture_year": 1992,
  "mileage_km": 88278,
  "auction_grade": "3.5",
  "interior_grade": "B",
  "has_repair_history": false,
  "one_owner": true,
  "equipment_list": ["PS", "PW", "AAC", "TV", "NAV", "ABS"]
}
```

15. Authentication & Authorization

15.1 Auth Flows

Google OAuth 2.0: User clicks Sign In with Google → redirected to Google consent screen → auth code returned → Auth Service exchanges for id_token → verifies token → upserts user record → issues JWT access + refresh tokens stored as httpOnly cookies.

Email/Password: User registers → email OTP verification sent → user verifies → account activated. Login issues JWT on correct credentials. Brute-force protection: 5 attempts per 10 minutes per IP via Redis rate limit counter.

15.2 JWT Claims

Claim	Value	Purpose
sub	user UUID	Subject identifier
tenant_id	tenant UUID	Tenant scope enforcement
email	user email	Display and audit
role	user tenant_admin super_admin	RBAC enforcement
jti	unique token ID	Token revocation tracking
exp	Unix timestamp	Token expiry (24 hours)

15.3 RBAC Permissions

Permission	super_admin	tenant_admin	user	api_client
Cross-tenant access	YES	NO	NO	NO
Manage tenant users	YES	YES	NO	NO
Upload and view analyses	YES	YES	YES	YES
Manage billing	YES	YES	NO	NO
Access admin portal	YES	Partial	NO	NO
Manage glossary	YES	NO	NO	NO

16. Payment Flow

16.1 PayHere Integration

PayHere is a CBSL-licensed Sri Lanka payment gateway. The integration uses the Hosted Payment Page (HPP) model: the platform creates an order, generates a signed PayHere payload, redirects the user to PayHere's checkout page, and receives an asynchronous IPN (Instant Payment Notification) webhook on payment completion.

16.2 Pricing Plans

Plan	Credits	Price (LKR)	Price / Analysis
Single Standard	1	250	LKR 250
Single Priority	1	500	LKR 500
Starter Pack	10	2,000	LKR 200
Professional Pack	50	8,500	LKR 170
Dealer Basic (monthly)	100/month	12,000/month	LKR 120
Dealer Pro (monthly)	300/month	25,000/month	LKR 83
Enterprise	Unlimited	Custom	Custom

16.3 Security — IPN Verification

All PayHere IPN notifications are verified using MD5 hash signature: `merchant_id + order_id + formatted_amount + currency + status_code + MD5(store_secret)`. If hash mismatch is detected, the webhook is rejected and the event is logged to the audit trail.

17. SaaS Multi-Tenant Design

17.1 Tenancy Model

Phase	Model	Isolation Level
Phase 1–3	Shared database, shared schema with tenant_id columns	Row-level security (RLS) via PostgreSQL policies
Phase 4 (Standard)	Shared database, schema-per-tenant on request	Schema isolation; dedicated connection pool
Phase 4 (Enterprise)	Dedicated RDS instance per enterprise tenant	Full physical isolation; separate backups

17.2 Tenant Routing

Tenant identity is extracted from incoming requests in priority order: (1) tenant_id claim in JWT, (2) subdomain resolution from Host header (e.g., dealer-name.auctioninsight.lk → tenant lookup). The resolved tenant_id is stored in TenantContext (thread-local) and applied to all database queries via RLS.

17.3 White-Label Features

Configurable Element	Options
Company name	Free text
Logo URL	PNG/SVG hosted on S3 CDN
Primary / secondary colour	Hex colour codes
Branded report watermark	Free text overlaid on report pages
Custom domain	CNAME or dedicated subdomain
Feature toggles	3D viewer, comparison, AI chat, public share links

18. Frontend Module Design

18.1 Module Hierarchy

The React application is organised into feature modules, shared components, services, and state management slices:

Module	Key Components
auth/	LoginPage, RegisterPage, GoogleOAuthCallback, useAuth hook
dashboard/	DashboardPage, AnalysisList, CreditWidget, QuickUploadCard
upload/	UploadPage, FileDropzone, UploadProgress, useUpload hook
analysis/	AnalysisViewPage → tabs (Overview, Condition, Damage, Comments, RawText)
analysis/damage/	DamageDiagram2D, DamageViewer3D (Phase 2), DamageMarker
payment/	CreditStorePage, PricingPlans, PayHereCheckout, PaymentHistory
reports/	ReportDownloadButton, ReportPreview, ShareReportModal
comparison/	ComparisonPage, ComparisonGrid — Phase 3
admin/	AdminLayout, UserMgmt, AnalysisQueue, ConfidenceReview, BillingOverview, GlossaryManager
tenant/	TenantSettings, BrandingEditor, APIKeyManager — Phase 4

18.2 State Management

Redux Toolkit manages global state across three slices: authSlice (current user, JWT, role), analysisSlice (current analysis, status updates, result data), and tenantSlice (white-label configuration). Real-time analysis progress is pushed via WebSocket and dispatched directly to analysisSlice.

18.3 Upload UX Flow

States: Idle (dropzone visible) → File selected → Validating (MIME/size) → Uploading (progress bar) → Queued (spinner) → OCR stage (35%) → Translation (60%) → Extraction (80%) → Report generation (95%) → Complete (redirect to result tab view).

19. Backend Microservices

19.1 Core API Service — Key Controllers

Controller	Endpoints	Business Logic
AnalysisController	POST /analyses, GET /analyses/{id}, GET /analyses	Upload validation, S3 upload, SQS dispatch, status queries
PaymentController	POST /payments/orders, POST /payhere/notify	PayHere order creation, IPN verification, credit allocation
UserController	GET /users/me, PATCH /users/me, DELETE /users/me/data	Profile, credit balance, PDPA erasure
ReportController	GET /analyses/{id}/report	Pre-signed S3 URL generation, PDF proxy
AdminController	GET /admin/analyses, POST /admin/analyses/{id}/review	Admin queue management, confidence review workflow

19.2 AI Orchestration — Pipeline Runner

The FastAPI AI Orchestrator picks up SQS messages, validates the job payload, and runs the analysis pipeline as a series of sequential async HTTP calls to the individual AI microservices. Each stage updates the analysis status in PostgreSQL and pushes a WebSocket event to the connected frontend client.

19.3 Idempotency Design

Every SQS message includes a deduplication_id equal to the analysis_id. If an analysis job is redelivered (e.g., after a Lambda timeout), the orchestrator checks the current status in PostgreSQL. If status is already 'complete' or 'failed', the duplicate message is discarded without re-processing.

20. AI Confidence Scoring

20.1 Confidence Formula

Each extracted field receives a composite confidence score from four sources, weighted to reflect their relative reliability:

```
field_confidence = (0.25 × ocr_confidence)
                  + (0.25 × translation_confidence)
                  + (0.30 × extraction_confidence)
                  + (0.20 × validation_confidence)
```

20.2 Validation Rules

Field	Validation Rule
mileage_km	0 < value < 999,999
displacement_cc	Must be a standard JDM engine size (660, 1000, 1300, 1500, 1600, 1800, 2000, 2400, 2500, 3000, 3500, 4000)
auction_grade	Must be in: S, 6, 5, 4.5, 4, 3.5, 3, 2, 1, RA, R, 0
manufacture_year	1980 <= year <= 2026
chassis_number	Length >= 6; alphanumeric (dashes allowed)
doors	Must be 2, 3, 4, or 5

20.3 Review Thresholds

Confidence Range	Action
>= 0.85	Auto-approve; report generated and delivered immediately
0.70 – 0.84	Report generated with low-confidence warning banner displayed
< 0.70	Analysis held; mandatory human review via admin portal; user notified (SLA: 2 hours)
Any single field < 0.50	Field marked UNVERIFIED with red badge in generated report

21. Error Handling Strategy

Error Category	HTTP Status	Auto-Retry	User Action
Validation error	422	No	Fix input and resubmit
Unauthenticated	401	No	Re-authenticate
Unauthorised	403	No	Check permissions
Rate limit exceeded	429	Yes (client)	Wait; Retry-After header provided
Insufficient credits	402	No	Purchase credits
OCR failure (async)	Status=failed	Yes — 3 retries	Re-upload clearer image if persists
LLM timeout (async)	Status=failed	Yes — 3 retries	Auto-resolved; notify if persistent
Storage error	503	Yes	System auto-retries

21.1 User-Facing Error Format

```
{
  "error": {
    "code": "INSUFFICIENT_CREDITS",
    "message": "You don't have enough credits for this analysis.",
    "detail": "1 credit required. You currently have 0.",
    "action": "Purchase credits to continue.",
    "action_url": "/credits"
  }
}
```

22. Security Architecture

22.1 Defence-in-Depth Layers

Layer	Controls Implemented
Network	AWS WAF (OWASP managed rules + rate limiting), CloudFront, Shield Standard DDoS protection
Transport	TLS 1.3 enforced; HSTS headers; certificate management via ACM
Application	JWT RS256, RBAC, CSRF tokens (SameSite cookies), input validation, SQL injection prevention (parameterised queries)
Data	AES-256 encryption at rest (RDS + S3 SSE); column-level encryption for PII fields
API	Rate limiting per IP and per user; request size limits; MIME sniffing validation
Infrastructure	VPC private subnets; IAM least-privilege roles; all secrets in AWS Secrets Manager
Audit	Immutable audit_log table; all sensitive actions logged with IP, user agent, and details

23. Compliance Considerations

23.1 Sri Lanka Personal Data Protection Act (PDPA) 2022

- Data minimisation: collect only data required for auction sheet analysis
- Purpose limitation: uploaded sheets used solely for analysis — not for model training without explicit opt-in
- User rights: erasure (DELETE /users/me/data) and portability (GET /users/me/data/export) endpoints
- Data retention: uploaded files 90 days; reports 1 year; deleted on request
- Breach notification: 72-hour notification to Data Protection Authority
- Data Processing Agreements (DPA) required for all dealer tenants

23.2 Payment Compliance

- PCI DSS: platform stores no raw card data; all card processing delegated to PayHere (PCI DSS Level 1)
- CBSL: PayHere licensed under Payment Devices, Systems and Settlement Act No. 28 of 2005

23.3 AI Transparency

All AI-generated reports carry the mandatory disclaimer: "This report is generated by artificial intelligence and should be used as a reference guide only. Auction Insight AI accepts no liability for purchase decisions made solely on the basis of this report. Always conduct a physical inspection before purchase."

24. Scalability Strategy

24.1 Traffic Pattern

Japanese auction activity peaks on auction days (typically Monday, Wednesday, Friday for USS). With Sri Lanka importers monitoring these events at UTC+5:30, expect 60–70% of daily analysis volume within a 4-hour window each auction day. The SQS-based pipeline absorbs burst traffic without dropping jobs.

24.2 Scaling Targets

Component	Phase 1	Phase 4
Concurrent analyses	50	500
API requests/second	50	500
DB connections	50	300 (via PgBouncer)
Storage	500 GB	10 TB

24.3 Auto-Scaling Policy

ECS Fargate AI worker tasks scale based on SQS queue depth. Target: 10 messages per task. Scale-out cooldown: 60 seconds. Scale-in cooldown: 300 seconds. Min capacity: 2. Max capacity: 20. This allows the platform to handle burst auction events without manual intervention.

25. DevOps & CI/CD

25.1 Pipeline Overview

Stage	Trigger	Actions
Feature branch push	Developer push	ESLint, Checkstyle, unit tests
Pull Request	PR opened	Integration tests, Docker image build
Develop merge	PR merged	Deploy to staging (ECS Fargate)
Staging deploy	Auto	E2E tests (Playwright), API contract tests
Release tag	Manual tag	Blue/Green deploy to production
Post-deploy	Auto	Smoke tests, monitor 10 min, auto-rollback if error rate > 1%

25.2 Infrastructure as Code (Terraform)

- All AWS resources managed as Terraform HCL modules
- Separate Terraform workspaces per environment (dev, staging, production)
- State stored in S3 with DynamoDB state locking
- Module structure: vpc, ecs, rds, elasticache, sqs, cloudfront, waf

25.3 Release Management

- Blue/Green deployments — zero downtime for all production releases
- Feature flags via AWS AppConfig for phased rollout of new capabilities
- Database migrations via Flyway — always tested on staging, never applied directly to production
- Secret rotation: AWS Secrets Manager auto-rotates RDS credentials every 30 days

26. Logging & Monitoring

26.1 Observability Stack

Concern	Tooling
Application metrics	AWS CloudWatch custom namespace metrics
Log aggregation	CloudWatch Logs (structured JSON, all services)
Distributed tracing	AWS X-Ray (trace IDs propagated across all service calls)
Alerting	CloudWatch Alarms → SNS → PagerDuty (Phase 1: email alerts)
APM dashboard	DataDog (Phase 2 onwards)

26.2 Key Alert Thresholds

Metric	Warning Threshold	Critical Threshold	Response
Analysis p95 latency	> 30 seconds	> 60 seconds	Scale AI workers
SQS queue depth	> 100 messages	> 500 messages	Auto-scale + alert on-call
OCR error rate	> 5%	> 15%	PagerDuty alert; check PaddleOCR logs
Payment failure rate	> 3%	> 10%	Alert finance team; check PayHere status
API 5xx error rate	> 1%	> 5%	Incident response; rollback if recent deploy
DB connection pool	> 70%	> 90%	Scale connections; check slow queries

27. Testing Strategy

27.1 Test Pyramid

Layer	Framework	Coverage Target	Focus
Unit tests	JUnit 5 (Java), pytest (Python)	80% line coverage	Business logic, extractors, validators, parsers
Integration tests	TestContainers (Java), pytest-asyncio	All API endpoints	Service APIs, DB operations, payment webhooks
E2E tests	Playwright (TypeScript)	20 key user journeys	Upload flow, payment, report download, admin review

27.2 AI Pipeline Regression Suite

A curated set of known auction sheet images (minimum 20 fixtures spanning USS, JAA, and JU formats) is maintained in the test fixture library. After any change to OCR configuration, glossary, or LLM prompts, the regression suite is run automatically. All fixtures must achieve > 95% field accuracy and > 0.80 per-field confidence before deployment is approved.

27.3 Acceptance Test Scenarios

ID	Scenario	Pass Condition
AT-001	Upload valid JPEG	Analysis completes < 35s; grade extracted correctly
AT-002	Upload multi-page PDF	Processed; first page used
AT-003	Upload blurry image	review_required=true; user notified
AT-004	PayHere success	Credits added within 5s of IPN receipt
AT-005	Insufficient credits	Upload blocked; clear error with action URL
AT-006	Tenant subdomain	Routes to correct white-label UI
AT-007	Admin review approval	User notified; status updated
AT-008	Rate limit exceeded	HTTP 429 with Retry-After header
AT-009	JWT expired	HTTP 401; refresh token flow triggered
AT-010	Report download	PDF contains all expected vehicle data fields

28. Deployment Architecture

28.1 AWS Infrastructure Summary

Component	Service / Config
Region	ap-south-1 (Mumbai) — lowest latency to Sri Lanka
Network	VPC 10.0.0.0/16; Public subnets (ALB, NAT); Private subnets (ECS, RDS, Redis)
CDN / Edge	CloudFront (SPA + static assets); AWS WAF; Route 53 wildcard DNS
Compute	ECS Fargate — all microservices; 0.5–4 vCPU, 1–8 GB per task; no EC2 management
Database	RDS PostgreSQL 15 Multi-AZ (db.t3.medium Phase 1); pgvector extension enabled
Cache	ElastiCache Redis 7 (cache.t3.small)
Messaging	AWS SQS Standard Queue for analysis job dispatch
Storage	S3 (uploads bucket + reports bucket); versioning and lifecycle policies enabled
Email	AWS SES for transactional emails (verification, notifications, receipts)
Secrets	AWS Secrets Manager for all credentials; auto-rotation enabled

28.2 Phase 1 Monthly Cost Estimate

Service	Configuration	Est. USD/month
ECS Fargate	5 services, 2–4 tasks average	\$120
RDS PostgreSQL	db.t3.medium Multi-AZ	\$95
ElastiCache Redis	cache.t3.small	\$25
S3 + Data Transfer	500 GB storage + egress	\$15
CloudFront	100 GB transfer	\$12
ALB	Standard configuration	\$20
SQS + CloudWatch	5M messages + logs/metrics	\$22
Route 53 + SES + Misc	DNS + email	\$5
AWS Subtotal		~\$314
OpenAI API	3,000 analyses × \$0.027	~\$81
PayHere	1.9% + LKR 25 per transaction	Variable

Total Platform Cost		~\$395 + payment fees
---------------------	--	-----------------------

29. Cost Optimization Strategy

- ECS Fargate Spot for AI worker tasks — 60–70% cost savings; tolerable for async pipeline (retry handles interruptions)
- S3 Intelligent-Tiering — automatically moves uploaded files to cheaper storage tiers after 30 days of inactivity
- OpenAI prompt caching — system prompts cached after first call; approximately 15% reduction in input token costs
- OCR batching — all pages of a multi-page PDF processed in one PaddleOCR invocation
- Redis TTL discipline — aggressive expiry prevents ElastiCache memory saturation and cluster upgrades
- Reserved Instances — purchase 1-year RDS reserved instance after Phase 1 validates baseline workload (saves ~40%)

30. Third-Party Service Evaluation

Service	Selected	Rationale
OCR Engine	PaddleOCR	Best Japanese accuracy; open source; runs in-container; no per-call cost
LLM	OpenAI GPT-4o	Superior structured JSON output; function calling; JSON mode; competitive pricing
Embeddings (RAG)	text-embedding-3-small	Cost-effective; same API ecosystem; 1536 dimensions suitable for pgvector
Payment (LKR)	PayHere	Dominant Sri Lanka gateway; CBSL licensed; best developer SDK; widely trusted
Payment (Intl)	Stripe	Best-in-class webhooks and developer experience; used as fallback
Vector DB	pgvector (PostgreSQL ext.)	Reuses existing RDS; no additional service; sufficient for Phase 1–4 scale
PDF Generation	WeasyPrint + Jinja2	Python-native; CSS-styled templates; no headless Chrome dependency
Email	AWS SES	Cheapest at scale; high deliverability from AWS IPs; native integration

31. 3D Rendering Strategy (Phase 2)

31.1 Vehicle Model Library

Type	Model File	Typical Japanese Vehicles Matched
Sedan	generic_sedan.glb	Skyline, Cefiro, Crown, Accord, Camry
Hatchback	generic_hatchback.glb	Corolla Hatch, Civic, Fit, Note, Swift
SUV	generic_suv.glb	Land Cruiser, Harrier, RAV4, X-Trail, CR-V
Van / MPV	generic_van.glb	Hiace, Alphard, Serena, Stepwagon, Noah
Coupe	generic_coupe.glb	Supra, RX-7, Silvia, Integra, 180SX
Pickup	generic_pickup.glb	Hilux, D-Max, Navara

31.2 Damage Zone Colour Coding

Severity	Colour (Hex)	Three.js Emissive Intensity
Information (replaced)	#3B82F6 (Blue)	0.3
Minor	#22C55E (Green)	0.3
Moderate	#F59E0B (Amber)	0.4
Severe	#EF4444 (Red)	0.5

32. Vehicle Damage Mapping Logic

32.1 Damage Code Parsing Algorithm

The damage parser processes all text extracted from the damage notation section (left half of the sheet, rows corresponding to the bad_points and inspector_comments regions). Each code is matched against a regex pattern:

Pattern: `[A-Z]{1,2}[0-9]?[A-Z]{0,2}[1-3]?`

Examples: D3 (dent severity 3), W2 (wave severity 2), U1 (rust severity 1), E (exchanged), XX (severe)

32.2 Aggregate Severity Score Calculation

Severity	Score Contribution per Item
Information (E codes)	0 points
Minor (suffix 1)	3 points
Moderate (suffix 2)	8 points
Severe (suffix 3 or XX)	20 points
Structural damage flag	+25 bonus points

The aggregate score is capped at 100. Scores below 15 = Low, 15–35 = Moderate, 35–60 = Significant, above 60 = High.

33. Export & Reporting Module

33.1 PDF Report Structure

Page	Content
Cover	Platform/dealer branding, report title, analysis reference number, generation date
Vehicle Summary	Make/Model/Year/Grade (prominent), key specs table, one-owner and repair history badges
Condition Assessment	Grading scale with position indicator, interior grade, equipment checklist, good selling points
Damage Report	Aggregate severity score gauge, damage items table with colour coding, 2D diagram
Translated Comments	Full inspector comments, selling points, additional notes
AI Notes & Disclaimer	Per-field confidence table, low-confidence warnings, mandatory AI disclaimer

33.2 Auction Grade Reference Table

Grade	Condition Description	Purchase Recommendation
6	Almost new / showroom condition	Excellent
5	Near perfect, minimal cosmetic marks	Highly recommended
4.5	Very good, minor surface blemishes	Recommended
4	Good condition, noticeable minor cosmetic issues	Good purchase
3.5	Average condition, some dents and scratches	Acceptable with caution
3	Below average, multiple cosmetic issues	Caution advised
2	Poor condition, significant damage	High risk
1	Very poor / major damage present	Not recommended
RA	Repaired accident vehicle (declared)	Verify repair quality carefully
R	Undeclared accident vehicle	High risk — specialist inspection required
0 / S	Special / parts only	Do not import

34. Admin Portal

34.1 Portal Sections

Section	Key Capabilities
Dashboard	Live queue depth, processing rate, today's analyses/revenue/new users, service health indicators
Analysis Queue	All analyses with status filters; priority management; force-retry failed analyses; assign to human reviewer
Confidence Review	Queue of analyses with confidence < 0.85; side-by-side original image vs AI extraction; approve/edit/re-run; SLA 2 hours
User Management	Search/filter all users; adjust credit balances (with audit reason); suspend/activate accounts
Tenant Management	Create/configure tenants; set branding, features, billing plans; generate API keys (Phase 4)
Billing Overview	Revenue by day/week/month (LKR); payment success/failure rates; credit consumption; subscription status
Glossary Manager	Add/edit/delete Japanese-to-English mappings; bulk CSV import; version history
System Health	Real-time service status; API latency trends; SQS queue metrics; OpenAI quota usage
Audit Log	Immutable, searchable log; filter by user, action, tenant, date range

35. User Flows

35.1 New User — First Analysis

Step	Action	System Response
1	User lands on marketing page; clicks 'Analyse Free'	Marketing page loads
2	Registers via email or Google	Account created; 1 free trial credit awarded
3	Verifies email (email registration)	Account activated
4	Sees dashboard with credit balance	CreditWidget shows 1 credit available
5	Clicks 'New Analysis'; selects file	Dropzone displayed; file selected
6	File uploads	Progress bar; MIME/size validation
7	Analysis queued	WebSocket connection opened; progress begins
8	Progress stages shown	OCR → Translation → Extraction → Report (WebSocket events)
9	Analysis completes	Browser notification; redirect to result page
10	Reviews tabbed results	Overview, Condition, Damage, Comments panels displayed
11	Downloads PDF report	PDF streamed from S3 via signed URL
12	Prompted to buy more credits	Pricing plans modal displayed

35.2 Dealer Subscription Flow

- Login via Google SSO on tenant-branded subdomain
- Dashboard shows shared team analysis queue and credit pool
- Upload individual or batch of auction sheets
- Monitor real-time progress for all queued analyses
- Review completed analyses; spot-check extracted fields
- Share branded report link with customer (public share, no login required)
- Monthly subscription auto-renews; credits refresh on billing cycle date

36. Sequence Diagrams

36.1 Full Analysis Pipeline

The following describes the complete asynchronous sequence from file upload to report delivery:

- User uploads file via React frontend to Core API (POST /analyses)
- Core API validates file, stores to S3, deducts credit, enqueues SQS message, returns 202 + WebSocket URL
- Client opens WebSocket connection to receive real-time progress events
- AI Orchestrator picks up SQS message, downloads file from S3
- Runs OCR → Translation → Extraction → Damage interpretation stages sequentially
- Each stage completion pushes a WebSocket event (stage name, progress %)
- Report service generates PDF, stores to S3, updates analysis status in PostgreSQL
- WebSocket pushes 'complete' event; frontend redirects user to result page

36.2 Payment + Credit Flow

- User selects plan on Credit Store page
- Frontend calls POST /payments/orders; Core API creates pending payment record
- Frontend receives PayHere signed payload; redirects user to PayHere checkout
- User completes card payment on PayHere's hosted page
- PayHere sends asynchronous IPN webhook to notify URL
- Core API verifies MD5 hash; marks payment success; adds credits to user balance
- User redirected to success URL; frontend fetches updated credit balance

37. Future Enhancements

37.1 Phase 5 Capabilities

Feature	Description
AI Chat Assistant	Natural language queries over analysis history and vehicle knowledge base, backed by RAG pipeline
Predictive Pricing	ML model correlating auction grade, mileage, year with Sri Lanka import market prices
Fraud Detection	ML model detecting odometer inconsistencies, chassis number anomalies, concealed accident patterns
Mobile App	React Native iOS/Android app with camera-based sheet capture and auto-perspective correction
Auction Agent Network	Marketplace connecting Sri Lankan buyers with vetted Japanese agents; analysis as trust layer
VIN Decoder	Cross-reference chassis numbers with JAIA/JASPA specification and recall databases
Sinhala & Tamil UI	Full localisation for Sri Lankan domestic market languages
Bulk API Processing	Webhook-triggered batch ingest for agents processing 50–100 sheets per auction day

38. Risk Analysis

38.1 Technical Risks

Risk	Probability	Impact	Mitigation
OpenAI API outage	Medium	High	Fallback to GPT-4o-mini; cached results for known patterns
OCR accuracy on non-standard sheets	High	Medium	Human review queue; expand fixture library; admin override
PaddleOCR model regression	Low	High	Pin model version in Docker; fallback to AWS Textract
DB performance degradation	Medium	High	Read replicas; query optimisation; PgBouncer
S3 cost overrun	Medium	Low	Intelligent-Tiering; compression; lifecycle policies

38.2 Business Risks

Risk	Probability	Impact	Mitigation
Low initial user adoption	Medium	High	Free trial credits; dealer partner launch; agent outreach
PayHere integration issues	Low	High	Extensive sandbox testing; Stripe fallback
Competitor launch	Medium	Medium	Defensible moat: domain vocabulary, historical data, dealer lock-in
AI liability for bad purchase decisions	Medium	High	Mandatory disclaimer; confidence scores; human review workflow
PDPA compliance gap	Low	High	Pre-launch legal review; DPA agreements; user consent flows

38.3 Operational Risks

Risk	Probability	Impact	Mitigation
Key person dependency	Medium	High	Documentation; cross-training; on-call runbooks
AWS ap-south-1 outage	Low	High	Multi-AZ design; documented failover to ap-southeast-1

OpenAI pricing increase	Medium	Medium	Token optimisation; evaluate open-source LLMs in Phase 5
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39. Technical Constraints

39.1 Known Limitations

Constraint	Impact	Mitigation
Handwritten text accuracy	OCR accuracy degrades on handwritten Japanese comments and damage annotations	LLM post-processing of low-confidence regions; human review for flagged analyses
Auction sheet format variance	Older and minor regional auction house sheets use non-standard layouts not covered by Phase 1 templates	Phase 1 targets 85–90% of volume on modern standard formats; edge cases via human review queue
Image quality dependency	Mobile photos of physical sheets introduce blur, perspective, and lighting issues	Pre-processing pipeline handles most cases; prompt user to re-upload if confidence too low
OpenAI context limits	GPT-4o 128K token limit requires careful batch size management for multi-page PDFs	Process sheets individually; context-window monitoring per request
LKR payment ceiling	PayHere maximum single transaction LKR 500,000 limits large enterprise subscription billing	Enterprise subscriptions invoiced and settled via bank transfer
3D model licensing	Quality GLTF vehicle models require licensing or custom commissioning	Budget LKR 150,000–250,000 for 6–8 generic models; Sketchfab commercial licensing
Sri Lanka connectivity	Rural dealers with 3G-only connections may struggle with large file uploads	TUS resumable upload protocol for files > 5 MB

39.2 External Dependencies

Dependency	SLA	Fallback Strategy
OpenAI GPT-4o	99.9%	Fallback to GPT-4o-mini; cached common responses
PayHere	99.5%	Manual bank transfer invoicing for enterprise; Stripe for international
AWS ap-south-1	99.99%	Multi-AZ deployment; failover runbook to ap-southeast-1

PaddleOCR (in-container)	N/A (self-hosted)	Pin Docker image version; AWS Textract as secondary
Google OAuth	99.9%	Email/password authentication always available as fallback

40. Estimated Development Roadmap

40.1 Phased Delivery Schedule

Phase	Scope	Duration	Target Launch
Phase 1	Core analysis engine, auth, payment, dashboard, reports, admin	3 months	Month 3
Phase 2	2D/3D damage visualisation, vehicle type detection	2 months	Month 5
Phase 3	Multi-sheet comparison, AI risk summary, PDF/Excel export, history	2 months	Month 7
Phase 4	Multi-tenant, white-label portals, public API, subscriptions	3 months	Month 10
Phase 5	RAG pipeline, AI chat, risk scoring, predictive analytics	3 months	Month 13

40.2 Phase 1 Sprint Breakdown

Sprint	Weeks	Deliverables
Sprint 1	1–2	AWS infrastructure (Terraform), DB schema (Flyway migrations), auth service, CI/CD pipeline
Sprint 2	3–4	File upload service, S3 integration, OCR microservice (PaddleOCR), image pre-processing
Sprint 3	5–6	Translation service, extraction service, LangChain pipeline, glossary v1 (500 terms)
Sprint 4	7–8	Damage interpreter, confidence scoring, WeasyPrint PDF report generation
Sprint 5	9–10	React frontend: auth, upload, dashboard, analysis result view, WebSocket status integration
Sprint 6	11–12	PayHere payment integration, admin portal v1, QA regression suite, staging hardening, launch prep

40.3 Core Team Composition

Role	FTE	Phase 1 Focus
Solution Architect / Tech Lead	1.0	Architecture, AI pipeline design, technical decision authority, code review

Backend Engineer (Java)	1.0	Spring Boot services, authentication, billing, REST API, SQS integration
AI / ML Engineer (Python)	1.0	PaddleOCR, LangChain orchestration, FastAPI microservices, WeasyPrint reports
Frontend Engineer (React)	1.0	React SPA, Tailwind UI, WebSocket integration, dashboard, upload flow
DevOps / Cloud Engineer	0.5	AWS, Terraform, CI/CD pipelines, CloudWatch monitoring, security configuration
QA Engineer	0.5	Test automation (JUnit/pytest/Playwright), acceptance testing, regression suite

40.4 Technology Stack Summary

Layer	Technology
Frontend	React 18, TypeScript, Tailwind CSS, Redux Toolkit, Three.js (Phase 2)
Backend — Core	Java 21, Spring Boot 3.x, Spring Security, JPA / Hibernate, HikariCP
Backend — AI	Python 3.12, FastAPI, LangChain 0.3, PaddleOCR, WeasyPrint, Jinja2
Database	PostgreSQL 15 (RDS Multi-AZ), pgvector extension, Redis 7 (ElastiCache)
AI / LLM	OpenAI GPT-4o, text-embedding-3-small (Phase 5)
Cloud Platform	AWS: ECS Fargate, RDS, S3, SQS, CloudFront, WAF, API Gateway, SES, Route 53
Infrastructure as Code	Terraform (modularised; separate workspaces per environment)
CI/CD	GitHub Actions (lint → unit → integration → deploy → E2E → smoke)
Containerisation	Docker, Amazon ECR
Monitoring	AWS CloudWatch, X-Ray; DataDog (Phase 2)

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